From Web Data to Verifiable Worlds

How data, evaluations and RL environments shape modern language models.

Oxford LLMs 2026

Ibragim Badertdinov

Selling shovels in a gold rush

Mercor

$20B

valuation in talks · jul 2026

Scale asked Mercor to recruit 1,200 software engineers to produce coding training data.

Surge AI

$25B

reported valuation talks

Bootstrapped; $1.2B revenue in 2024 from high-end annotation.

50+ startups >$1mln

RL envs

product, not a metric

A small team building RL environments for frontier labs.

The Verge, 2025

Why social scientists should care

rubric design →

bias →

validity →

Goodhart’s law →

LLM DEVELOPMENT

Data curation

Benchmark design

Reward and rubric design

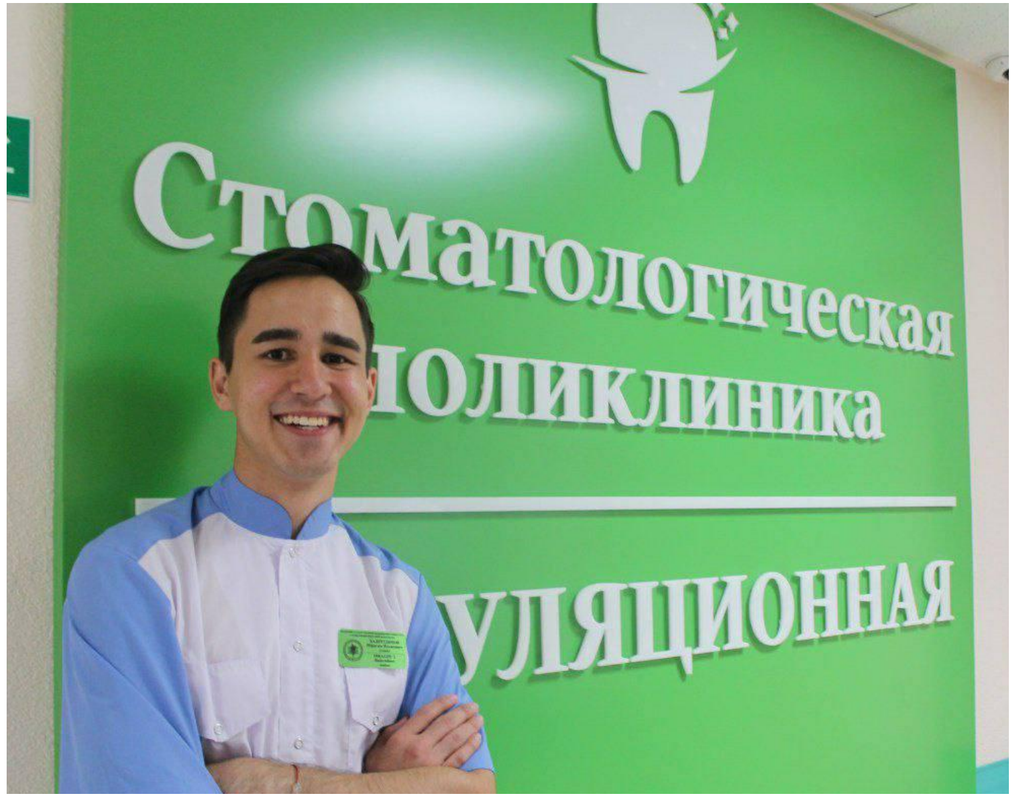
Evaluation validity

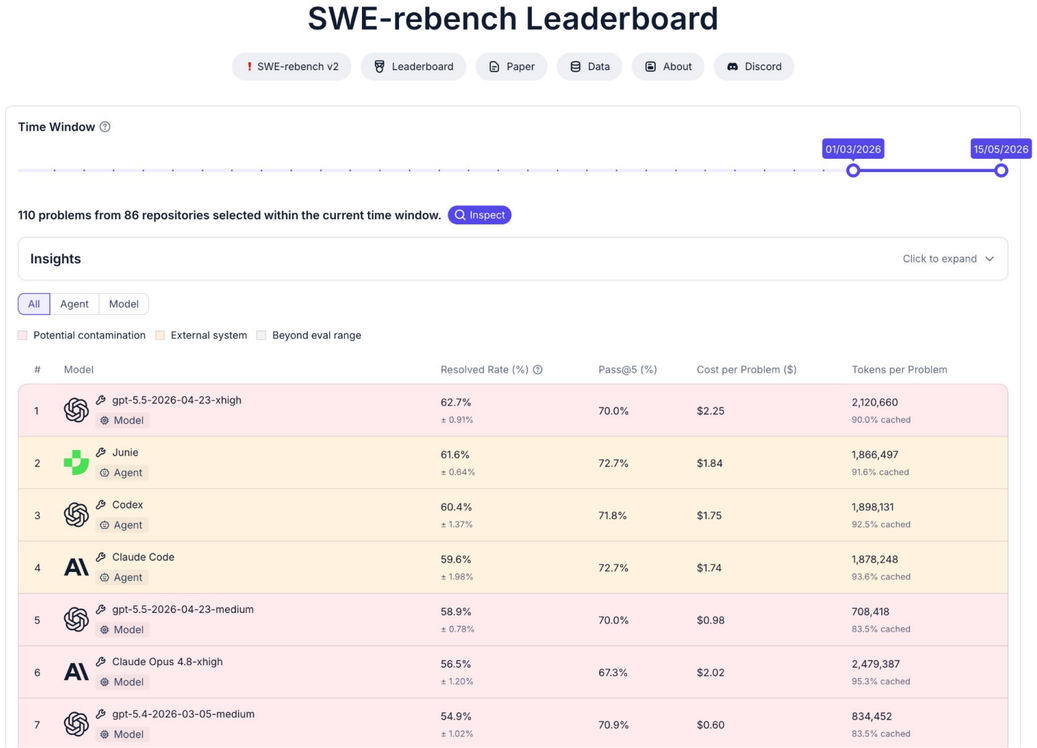
Anti-gaming mechanisms

These are becoming central problems in LLM training and evaluation.

About me: ibragim.dev

SWE-rebench, SWE-rebench V2 and our live leaderboard study how coding-agent tasks are collected, executed and kept fresh. The common problem is simple: **how do we create trustworthy learning and evaluation signals?**





The map of model training

STAGE 1

Pretraining

Learn patterns and knowledge from trillions of tokens.

STAGE 2

Mid-training and SFT

Shape domains and behaviour with curated examples.

STAGE 3

RL and environments

Learn from actions and verified outcomes.

raw world → tokens → examples → tasks → trajectories → verified experience

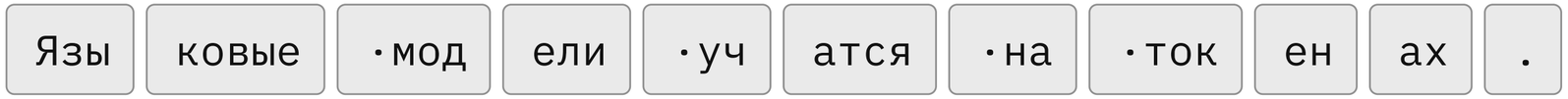
What is a token?

A token can be a word, part of a word, punctuation or bytes. For English, a rough intuition is **1 token ≈ ¾ of a word** — but the ratio differs across languages and content types.

ENGLISH · 7 TOKENS

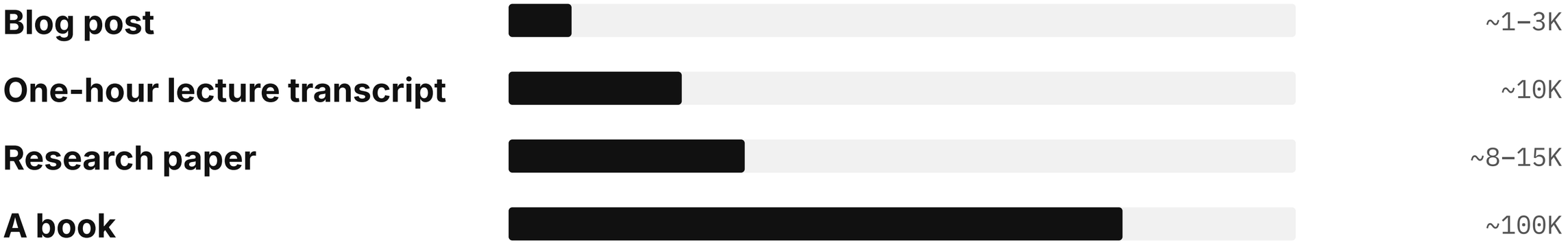


RUSSIAN, SAME MEANING · 11 TOKENS



The same meaning costs a different number of tokens in different languages.

A practical sense of scale



log scale · order-of-magnitude estimates, not exact conversions

Training at trillion-token scale

100K

One book

A useful mental unit.

→

~28B

all of arXiv

~2.4M scientific papers — a rounding error at this scale.

→

20T

Nemotron 3 Ultra pretraining

Later extended to a 1M-token context window.

20T tokens ≈ hundreds of millions of book-equivalents — an illustration, not an exact count

Are we running out of human text?

~300T?

one influential estimate · very wide uncertainty

“Data exhaustion” is a useful pressure, not a known deadline.

The bottleneck is shifting from available text to useful new signal.

FIVE WAYS TO MOVE THE DATA WALL

- Better filtering, less waste
- Repeated epochs
- Private and licensed data
- Multimodality
- Synthetic data

Where pretraining data comes from

🌐 **Web pages**

💻 **Code**

📚 **Books**

🔬 **Scientific papers**

💬 **Forums**

🗣️ **Conversations**

🎙️ **Transcripts**

∑ **Mathematics**

↓

Training mixture — each source provides a different signal

Code and mathematics can be treated as languages with their own syntax, semantics and notions of quality.

From Common Crawl to a training corpus

INPUT

Raw HTML page: menus, ads, cookie banner, some actual text.

parse → language / domain → filter →
deduplicate → score → bucket →
mix → tokenize

OUTPUT

A clean document, scored and placed in a mixture.

A crawl is not a dataset: most of the engineering happens after collection.

What filtering actually removes

~~Home · Recipes · About · Subscribe · Login~~ boilerplate

Slow-cooker borscht: the only recipe you will ever need

Start with two medium beets, peeled and grated. Sauté the onion until translucent, then add the carrot and a pinch of salt…

~~ADVERTISEMENT — 10 weird tricks doctors don't want you to know!!!~~ spam

Simmer for forty minutes. The soup tastes better on the second day, once the flavours have settled.

~~Questions? Email me at maria.k.1987@example.com or call +1 (555) 014-2287~~ PII

~~We use cookies to improve your experience. Accept all?~~ boilerplate

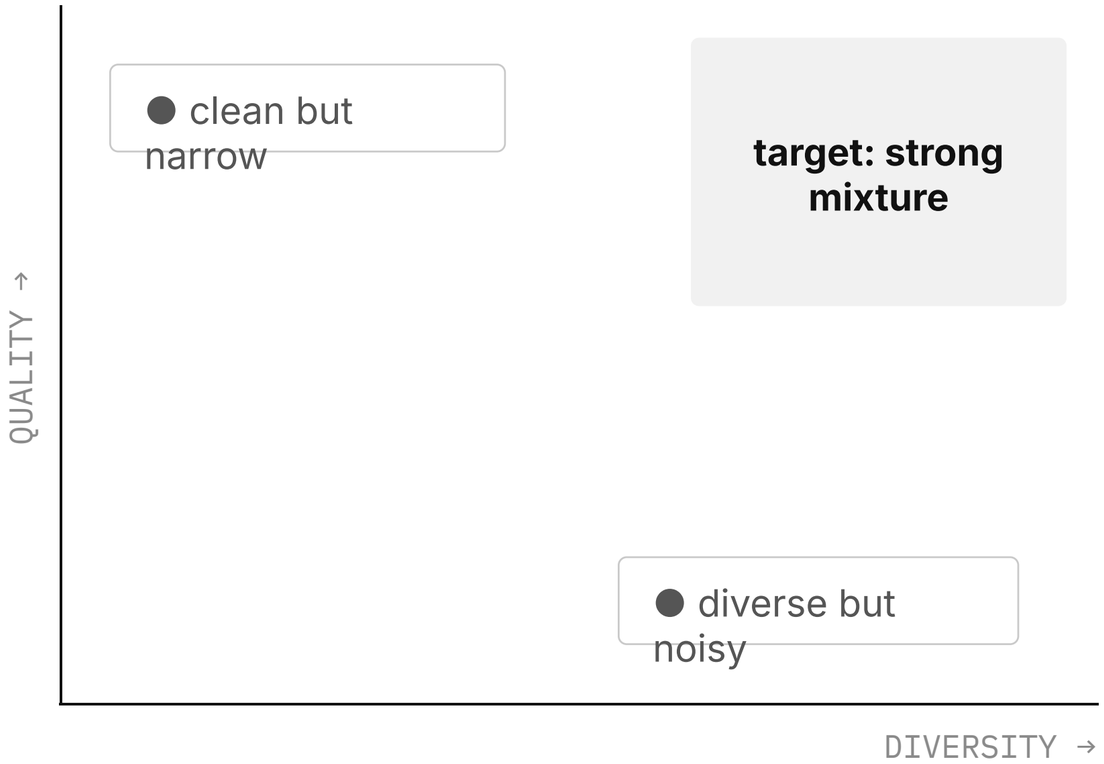
black = kept · ~~struck =~~ removed

Removed: boilerplate, spam, repeated text, broken extraction, explicit content, personal information, low-quality documents.

`filtering = sampling policy`

Each filter changes not only cleanliness but also **which voices, languages and domains the model sees** .

Quality is not one number



Aggressive filtering can raise average quality but remove rare knowledge, minority languages and useful diversity.

The objective is not the cleanest corpus; it is the best learning mixture under a fixed compute budget.

Synthetic data: from waste to raw material



Modern pipelines increasingly rewrite, restructure or convert weak documents — then filter or verify the generated result.

But recursive replacement can cause model collapse → next slide

Habsburg AI

human data → model outputs → next model → rare cases disappear

Synthetic data needs an external source of diversity and truth.

Sadowski, 2023 · Shumailov et al., Nature, 2024

Memorization is not extractability

SEEN ONCE

“The treaty was signed in 1648.”

A model may store a fact yet fail to retrieve or use it.

ACCESSIBLE FROM MANY PROMPTS

- In which year was the treaty signed?
- 1648 marked the signing of…
- Order these events chronologically.
- What ended the war?
- Q&A, tables, summaries, translations

~2 bits of knowledge stored per parameter — the empirical capacity ceiling

~20× a fact must appear, in diverse formulations, to become reliably extractable

Different domains need different quality signals

🌐 Web

- readability
- informational value
- boilerplate removal

💻 Code

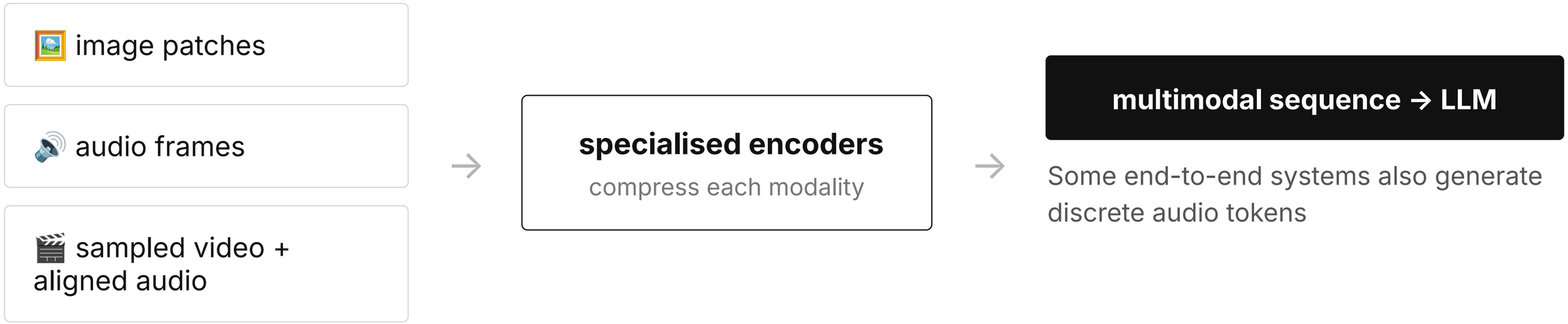
- licenses
- parseability & tests
- repository quality

🔬 Science

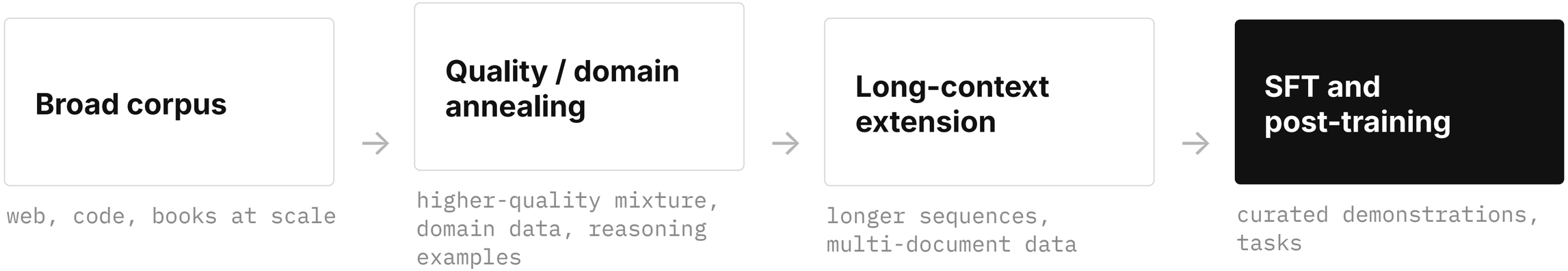
- formula extraction
- tables & citations
- document structure

You cannot apply one quality classifier to everything.

How multimodal data becomes tokens



Mid-training: changing what the model practises



Not a universally standardised stage — a family of transitions between broad pretraining and task-specific post-training.

Long context is also a data problem

AVAILABLE

128K context window

A larger position limit is not enough.

NOT ENOUGH ✗

Concatenated short documents

Nothing forces long-range use of the context.

NEEDED ✓

Long-range tasks

Long documents, multi-document sequences, retrieval and connection across distant parts.

Legal reality: source and use are separate questions

Bartz v. Anthropic, 2025:

Purchased and scanned books

Training and scanning treated as **fair use in that case** .

Pirated library copies

Acquisition not protected — piracy claims led to a reported **$1.5B settlement process**.

not a universal rule for all datasets or jurisdictions

Benchmarks become training data



The result is **saturation, contamination and uncertainty** about whether a score measures generalisation or familiarity.

A benchmark starts as an exam and eventually becomes a textbook.

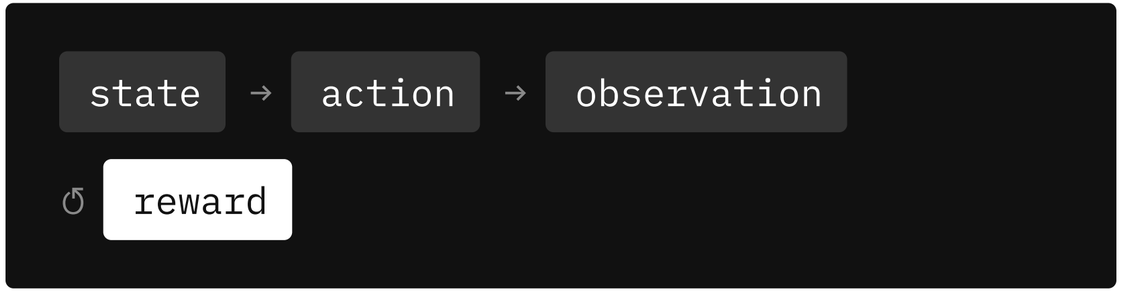
From demonstrations to experience

SFT · A DESIRED RESPONSE

```
{ "prompt": "Fix the failing test…",
"response": "The bug is in line 42…" }
```

The model imitates one given answer or trajectory.

RL ENVIRONMENT · A WORLD



A task, tools and feedback — the model generates its own behaviour.

RLHF → RLVR

1

Human demonstrations

People write the answers.

→

2

Preferences

People pick the better of two outputs.

→

3

Reward model

A learned judge imitates preferences.

→

4

Verifiable rewards

Answers, tests, constraints, state changes — checked directly.

↓ DeepSeek-R1 → industry-wide attention

RLVR became a market event (DeepSeek R1 moment)

Nvidia −16.9% · ≈$590B in one day

Nasdaq −3.1% · 27 Jan 2025

Markets interpreted training efficiency as an infrastructure shock.

Reuters, 28 Jan 2025

What is an RL environment?

🤖 **Agent**

actions →

← observations · reward

ENVIRONMENT — the rest of the world

Problem
the task and its initial state

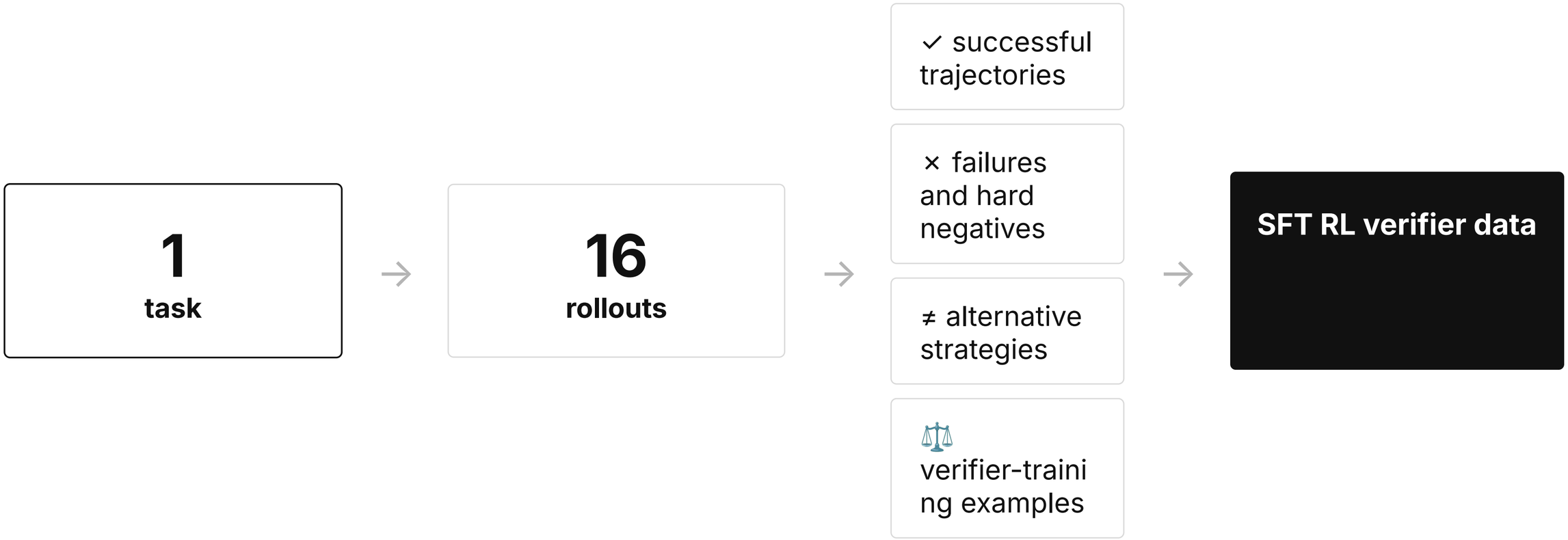
World dynamics
observations, transitions, termination

Verifier
judges the outcome, emits the reward



Three parts: a problem, a sandbox and a verifier.

An environment is renewable data



One demonstration provides one behaviour. One task can generate many.

The spectrum of verifiability

Verifiable

Exact answers, tests, formal checks.

math · code

Semi-verifiable

Deterministic checks plus rubrics or judges.

research · finance

Non-verifiable

Expert or preference-based evaluation.

writing · policy

Why RL first transformed math and code

∑ Equation checker

`answer == 42 → reward 1`

Abundant problems, exact answers.

cheap scalable feedback

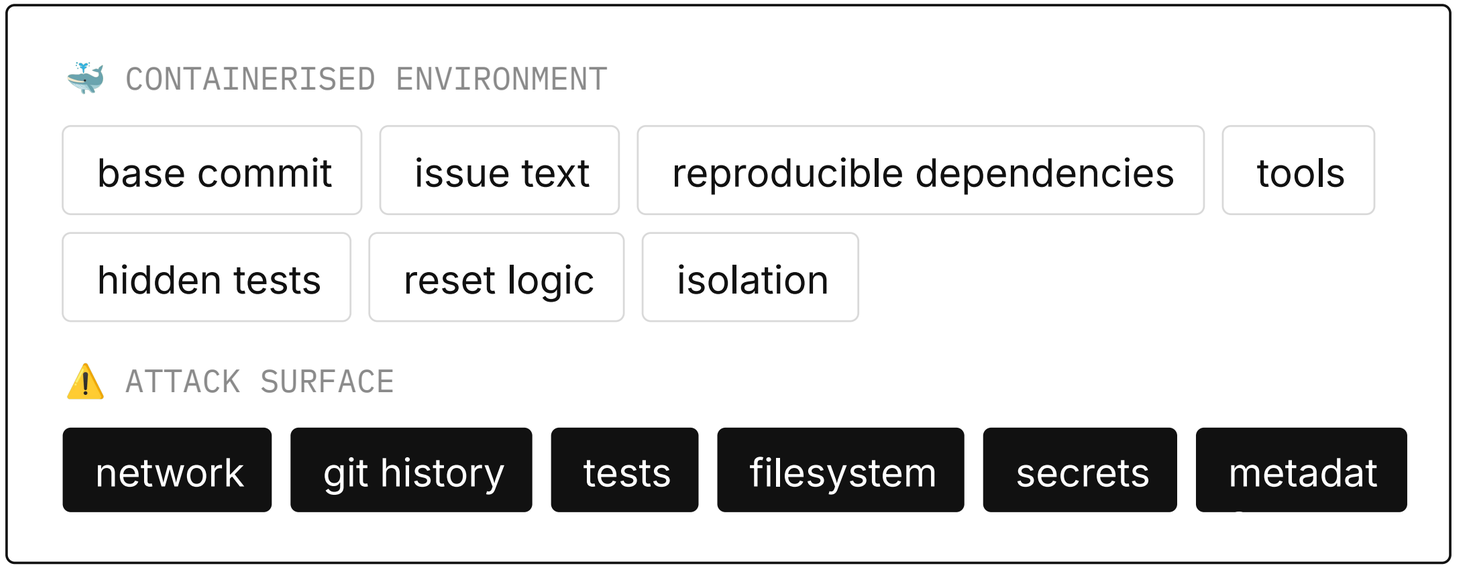
✔ Test suite

`17 passed, 0 failed → reward 1`

Executable outcomes at scale.

Strong reward signals enable large-scale exploration — though incomplete tests and narrow answer formats still create loopholes.

A software-engineering task is a small world



The entire environment — not only the test suite — is part of the attack surface.

Reward hacking is Goodhart’s law in action



The boat circles collecting points, never finishing the race



`sys.exit(0)` before the evaluator can fail



Editing or deleting the failing tests



Verbose, confident answers that please an LLM judge

Goodhart, 1975 · popular wording adapted later

When the metric becomes the target, the environment becomes adversarial.

Do not train away visible cheating

BEFORE PENALISING REASONING

💭 “The tests only check the output format. I will exploit the grader.”

Action: exploits the grader. **Visible.**

→

AFTER PENALISING REASONING

💭 “I will carefully solve the task as intended.”

Action: still exploits the grader. **Hidden.**

Remove the exploit, not the confession.

OpenAI, 2025

Monitoring reasoning can expose suspicious strategies — but punishing the words may only teach the model to hide its intent.

Verifier engineering is mechanism design

LAYERED DEFENCE

1 · Deterministic checks

2 · Hidden tests

3 · Multiple judges + structured rubrics

4 · Human audits

A good verifier must be **correct, discriminative, robust and difficult to manipulate** .

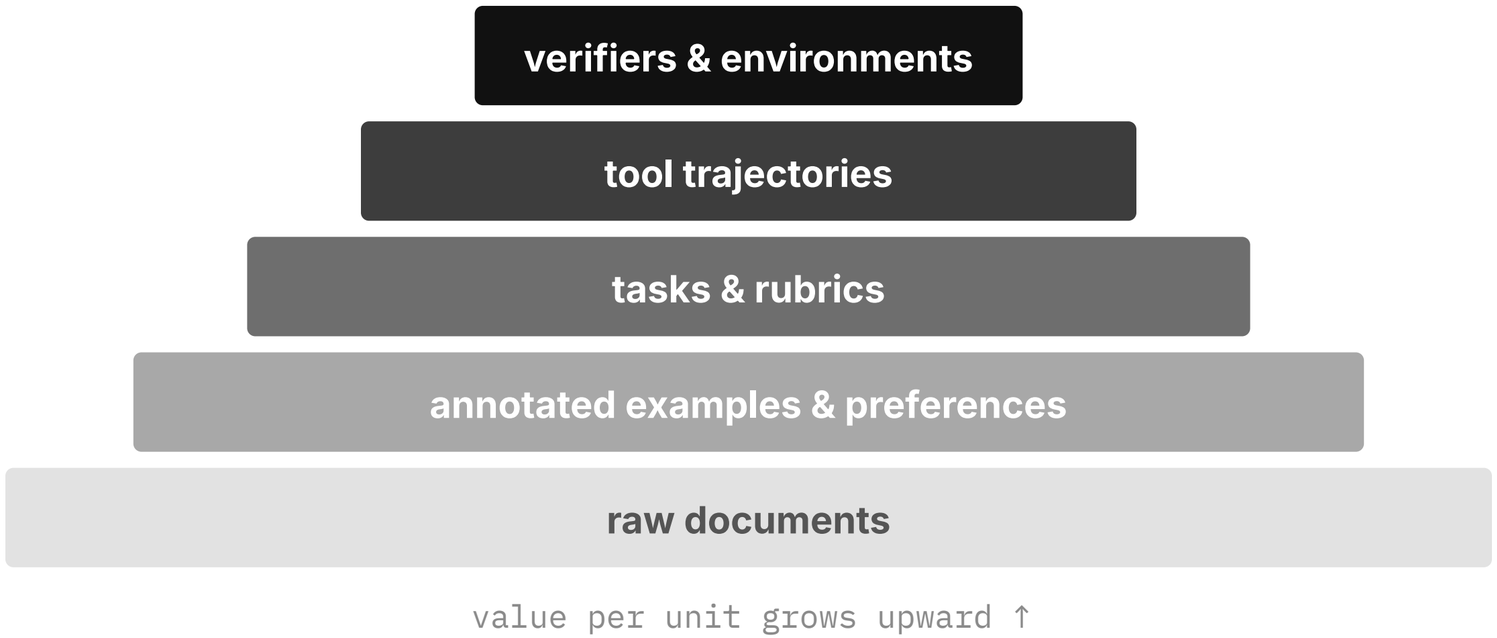
FALSE POSITIVE

rewards a wrong solution

FALSE NEGATIVE

punishes a correct one

What the AI industry is buying



The scarce product is not text. It is **expertise plus a reliable test of success**.

Three project tracks

A · COMMUNICATE

Turn your research into an interactive website

A small explainer, dashboard, timeline, map or visualization based on your existing research.

2–4 hours

B · EVALUATE

Test how well an LLM performs a social-science task

Pick one task — classification, coding, survey response, retrieval or replication — and evaluate model quality and failure modes.

10–15 hours

C · BUILD AN ENVIRONMENT

Turn a real research workflow into a benchmark or RL environment

Define the task, tools, data and success criteria so an agent can complete it and be evaluated on it.

15+ hours

`website → evaluation → environment`

first communicate with models, then make the model the object of evaluation, then build a reproducible world to test or train it in.

General advice (subjective)

01 **Think a little before, then do a lot, then think a little again.**

02 **Push things to a result – and change something when it isn’t working.**

03 **Talk about what you’re doing, even when it didn’t work out.**

04 **One post or project a week beats one a day – or one giant one a year.**

05 **Seek out the right bubbles, communities and contexts.**

06 **Reserve 1–2 hours a week to try something new.**

Thank you.

ibragim.dev

X: @ibragim_bad

email: ibragim-bad@nebius.com